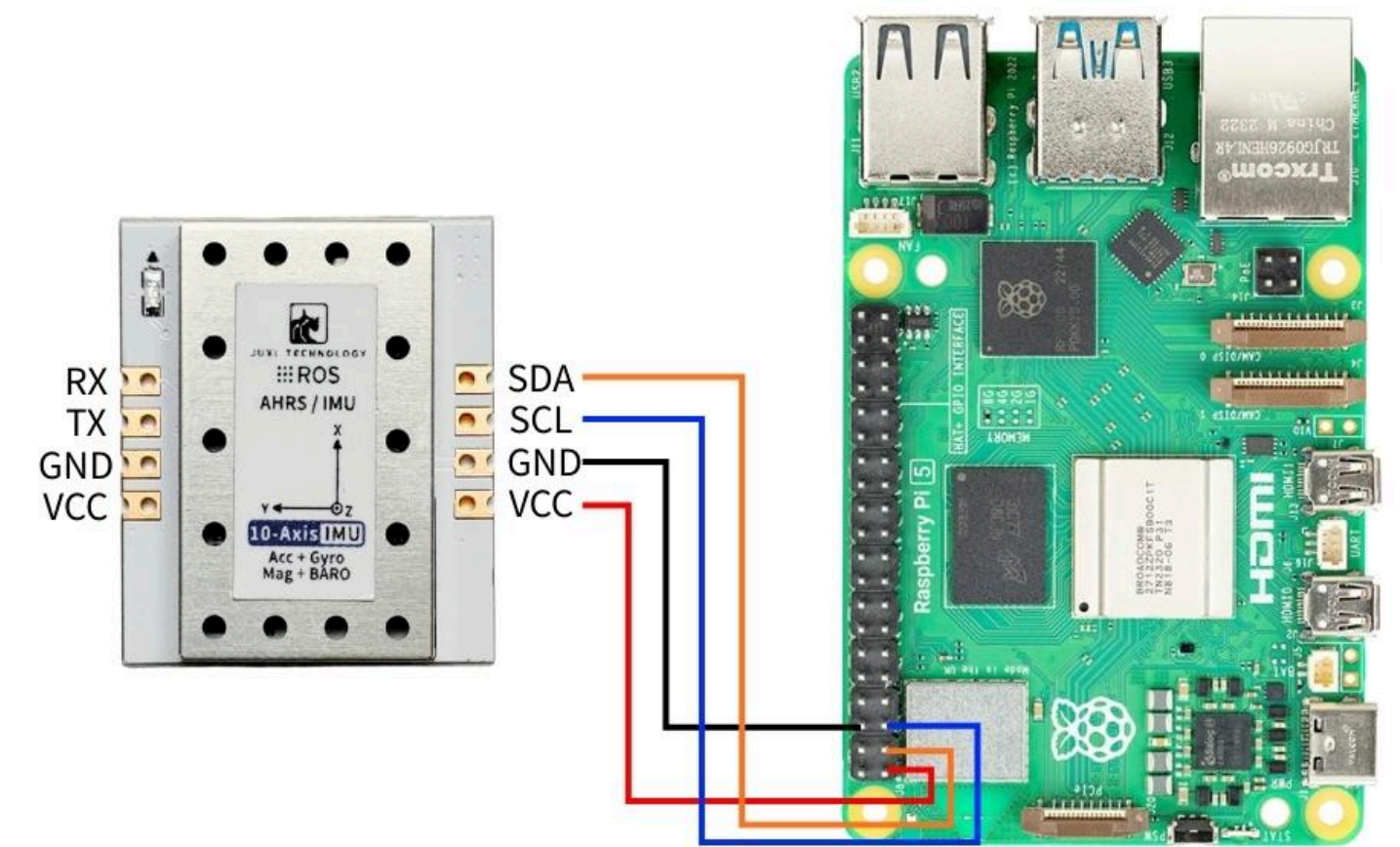


Raspberry Pi 5

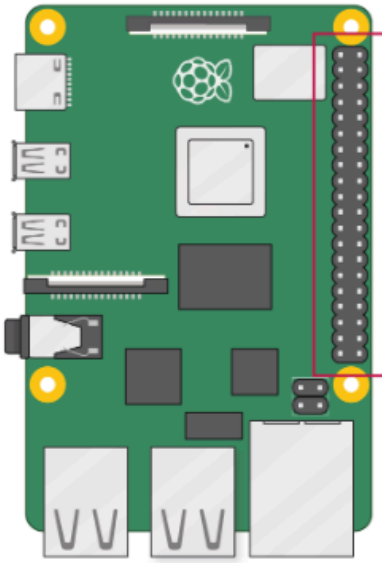
1. Connect the device

This tutorial takes the Raspberry Pi 5 motherboard and the official 64-bit version of the mirroring as an example.

Connect the IMU attitude sensor to the I2C interface of Raspberry Pi 5 as shown in the figure below.



IMU姿态传感器	树莓派5（物理引脚）
SDA	3
SCL	5
GND	GND
3.3V	3.3V



3V3 power	1	2	5V power
GPIO 2 (SDA)	3	4	5V power
GPIO 3 (SCL)	5	6	Ground
GPIO 4 (GPCLK0)	7	8	GPIO 14 (TXD)
Ground	9	10	GPIO 15 (RXD)
GPIO 17	11	12	GPIO 18 (PCM_CLK)
GPIO 27	13	14	Ground
GPIO 22	15	16	GPIO 23
3V3 power	17	18	GPIO 24
GPIO 10 (MOSI)	19	20	Ground
GPIO 9 (MISO)	21	22	GPIO 25
GPIO 11 (SCLK)	23	24	GPIO 8 (CE0)
Ground	25	26	GPIO 7 (CE1)
GPIO 0 (ID_SD)	27	28	GPIO 1 (ID_SC)
GPIO 5	29	30	Ground
GPIO 6	31	32	GPIO 12 (PWM0)
GPIO 13 (PWM1)	33	34	Ground
GPIO 19 (PCM_FS)	35	36	GPIO 16
GPIO 26	37	38	GPIO 20 (PCM_DIN)
Ground	39	40	GPIO 21 (PCM_DOUT)

2. Check device status

First, install I2Ctool, then enter the following in the terminal:

代码块

```
1 sudo apt-get update
2 sudo apt-get install -y i2c-tools
```

View I2C Devices

代码块

```
1 sudo i2cdetect -y -r -a 1
```

```
pi@pi: ~  
文件(F) 编辑(E) 标签(T) 帮助(H)  
pi@pi:~$ sudo i2cdetect -y -r -a 1  
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f  
00: -- -- -- -- -- -- -- -- -- -- -- -- -- --  
10: -- -- -- -- -- -- -- -- -- -- -- -- -- --  
20: -- -- -- 23 -- -- -- -- -- -- -- -- -- --  
30: -- -- -- -- -- -- -- -- -- -- -- -- -- --  
40: -- -- -- -- -- -- -- -- -- -- -- -- -- --  
50: -- -- -- -- -- -- -- -- -- -- -- -- -- --  
60: -- -- -- -- -- -- -- -- -- -- -- -- -- --  
70: -- -- -- -- -- -- -- -- -- -- -- -- -- --  
pi@pi:~$
```

3. Install the driver library

3.1 Install the Python libraries required for the code

代码块

```
1 sudo apt update  
2 sudo apt install -y python3-serial  
3 sudo apt install -y python3-smbus2
```

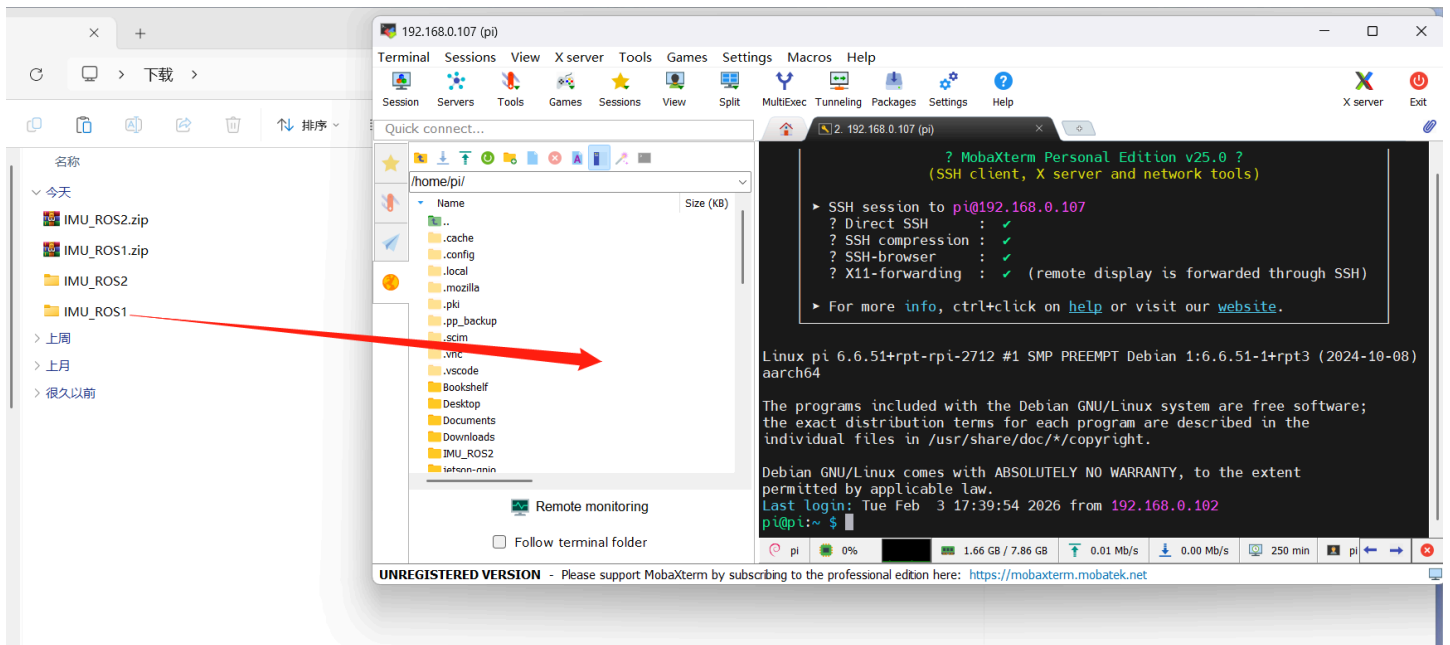
3.2 Transfer Files

 IMU_ROS2.zip
357.96KB



Friends who are not yet familiar with using MobaXterm to transfer files, please refer to the following webpage for detailed installation and operation methods of MobaXterm: [🌐 文件远程传输](#)

Drag the decompressed files onto Raspberry Pi 5 via MobaXterm software.

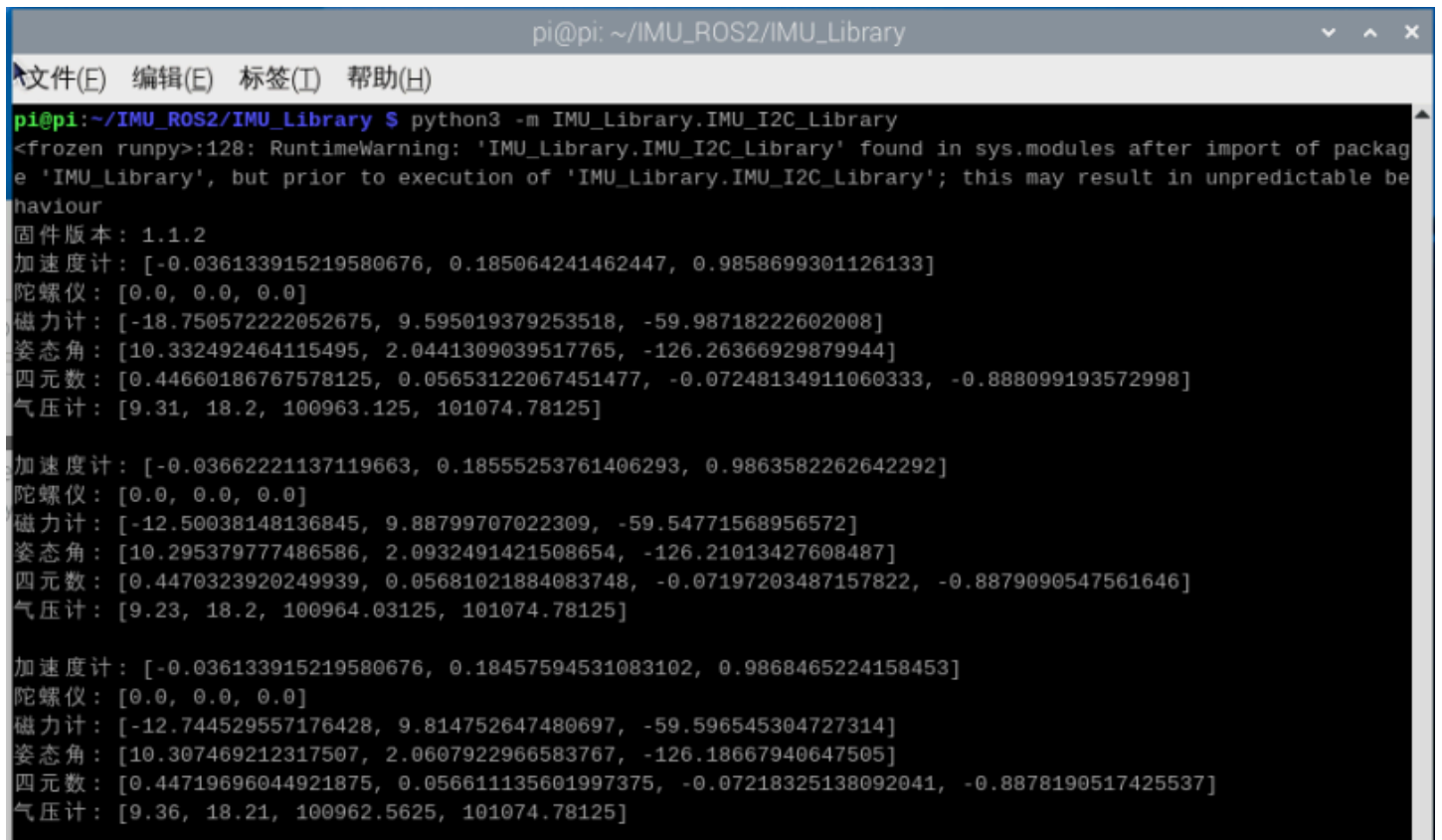


4. View IMU data

Enter the ~/IMU_Library directory and run the IMU_Serial_Library.py file

代码块

```
1 cd ~/imu_ros2/src/IMU_ROS1/IMU_Library
2
3 # 运行 IMU 数据打印文件
4 python3 -m IMU_Library.IMU_I2C_Library
```



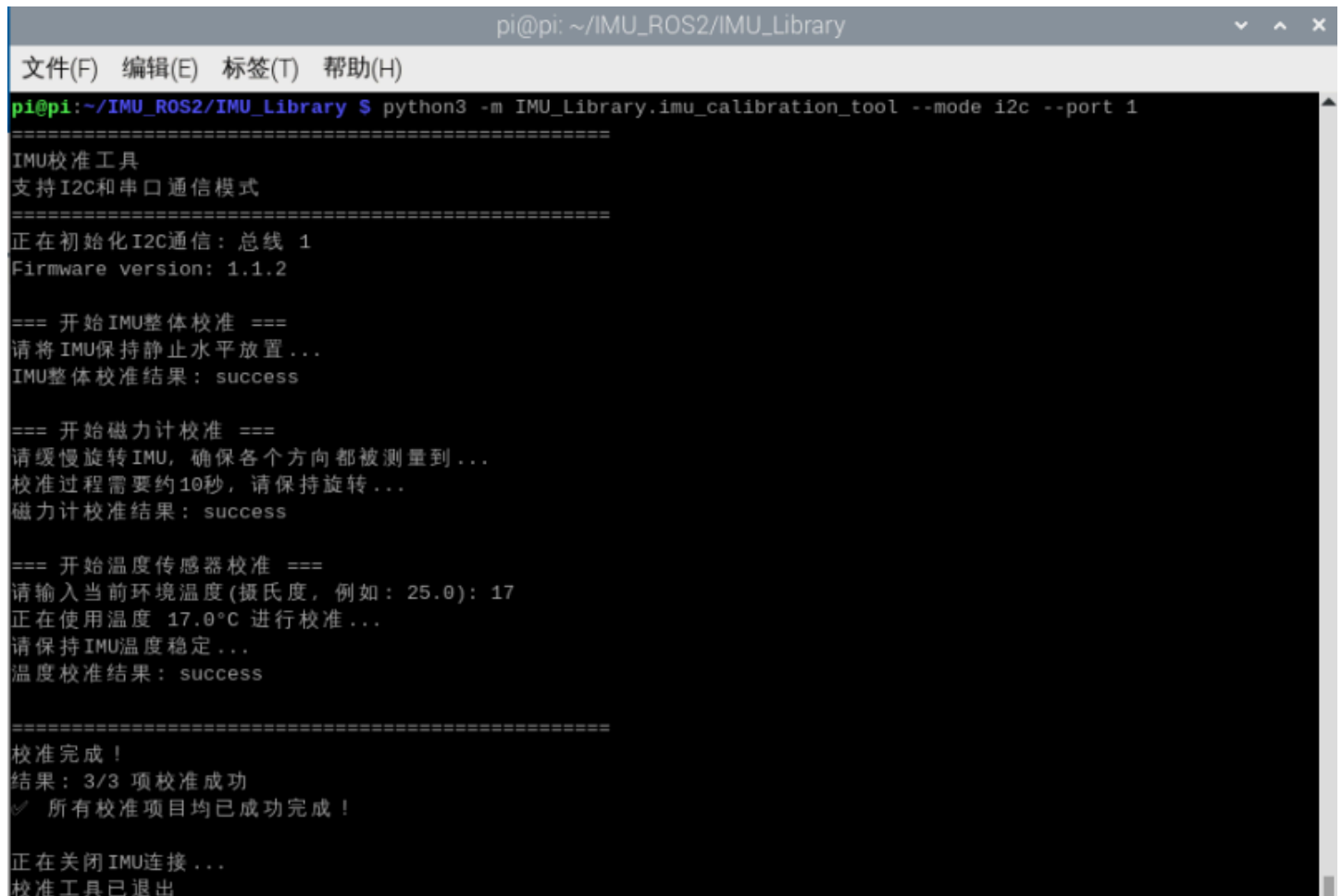
Note: The above is the data reading for a 10-axis IMU. The 6-axis has no Magnetometer and Barometer data, and the 9-axis has no Barometer data.

5. IMU Calibration

Enter the ~/IMU_Library directory and run the imu_calibration_tool.py file

代码块

```
1  cd ~/IMU_Library
2
3  # 运行 IMU 校准代码文件 --I2C通讯校准
4  # 执行所有校准（整体、磁力计、温度）
5  python3 -m IMU_Library.imu_calibration_tool --mode i2c --port 1
6
7  # 仅整体校准
8  python3-m IMU_Library.imu_calibration_tool --mode i2c --port 1 --calibrate imu
9
10 # 仅磁力计校准
11 python3-m IMU_Library.imu_calibration_tool --mode i2c --port 1 --calibrate mag
12
13 # 仅温度校准
14 python3-m IMU_Library.imu_calibration_tool --mode i2c --port 1 --calibrate temp
```



```
pi@pi: ~/IMU_ROS2/IMU_Library
文件(F) 编辑(E) 标签(T) 帮助(H)
pi@pi:~/IMU_ROS2/IMU_Library $ python3 -m IMU_Library.imu_calibration_tool --mode i2c --port 1
=====
IMU校准工具
支持I2C和串口通信模式
=====
正在初始化I2C通信：总线 1
Firmware version: 1.1.2

=== 开始IMU整体校准 ===
请将IMU保持静止水平放置...
IMU整体校准结果：success

=== 开始磁力计校准 ===
请缓慢旋转IMU，确保各个方向都被测量到...
校准过程需要约10秒，请保持旋转...
磁力计校准结果：success

=== 开始温度传感器校准 ===
请输入当前环境温度(摄氏度，例如：25.0)：17
正在使用温度 17.0°C 进行校准...
请保持IMU温度稳定...
温度校准结果：success

=====
校准完成！
结果：3/3 项校准成功
✓ 所有校准项目均已成功完成！

正在关闭IMU连接...
校准工具已退出
```

6. Precautions

Raspberry Pi 5 needs to enable the i2c pin in advance.

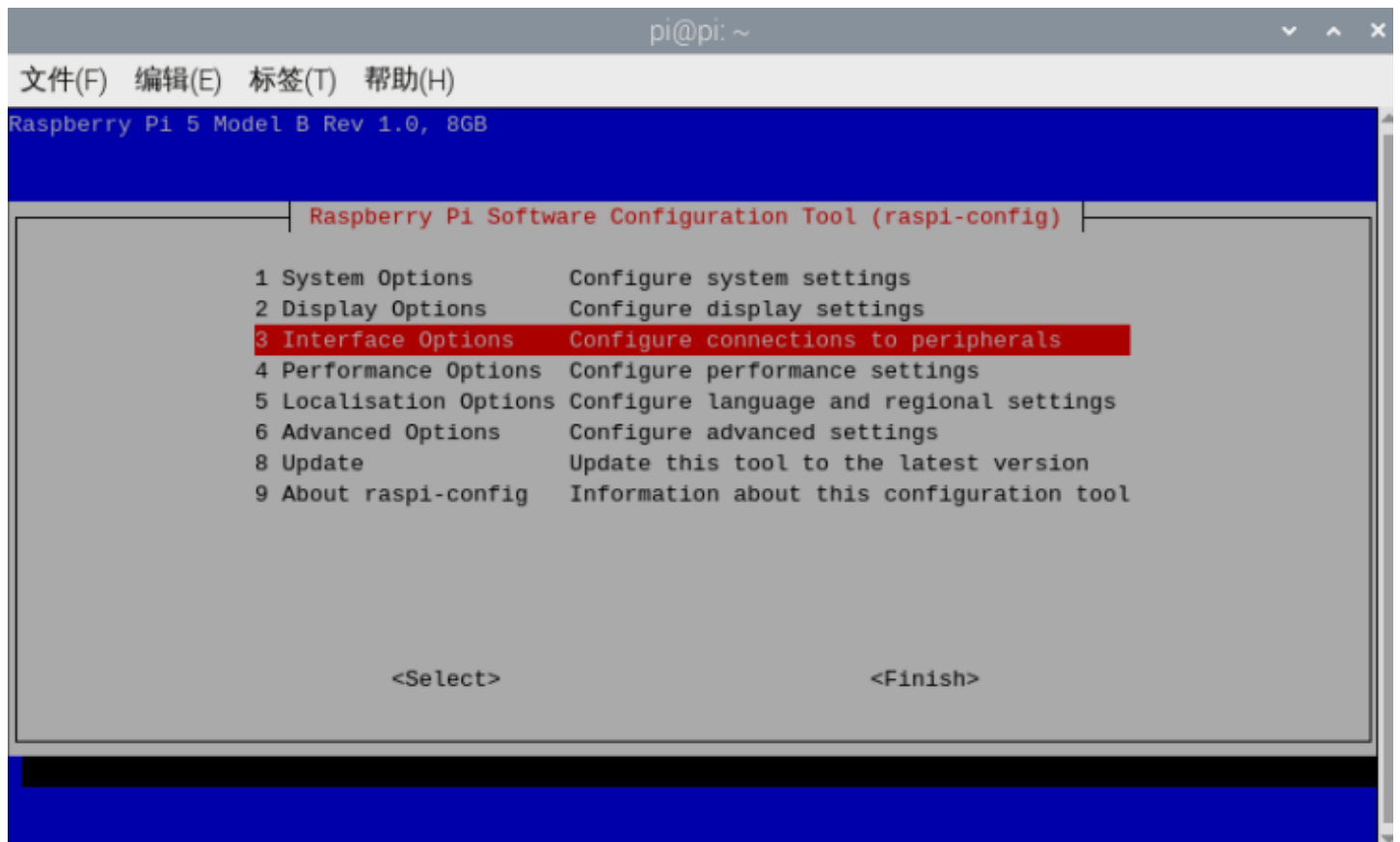
The opening operation is as follows:

Terminal run command

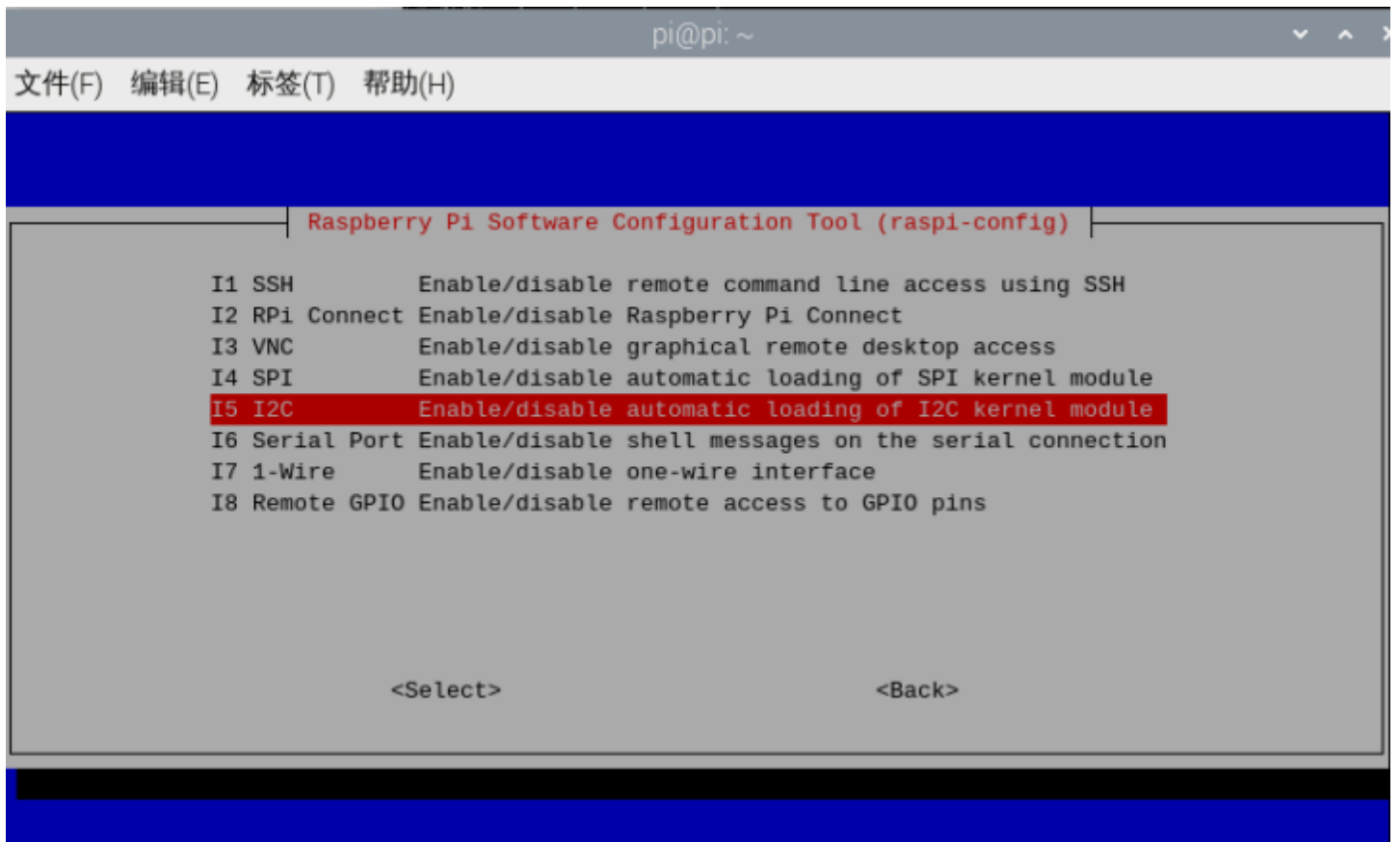
代码块

```
1 sudo raspi-config
```

Select using the arrow keys on the keyboard, and press the Enter key on the keyboard to enter after selection



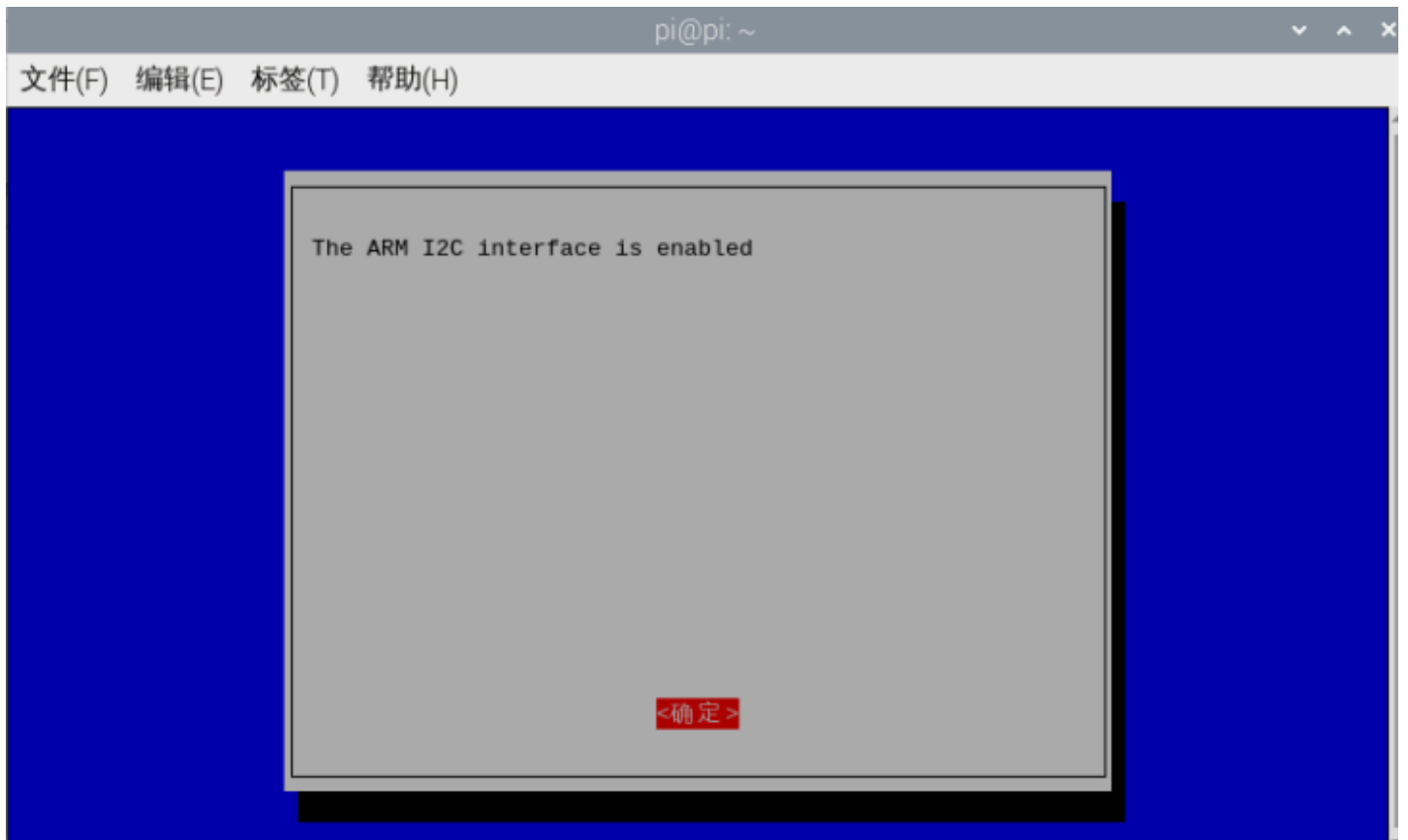
Select I2C, and after selection, press the Enter key on the keyboard to enter.



After selecting I2C, press the Enter key on the keyboard, use the arrow keys to select Yes, and then press the Enter key to confirm.



Press Enter to confirm



Press the arrow keys to select Finish, then press Enter to exit the configuration.

